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Research IT

High Performance Computing Update

CDT Journal Club, April 2025

Dr Patricia Ternes

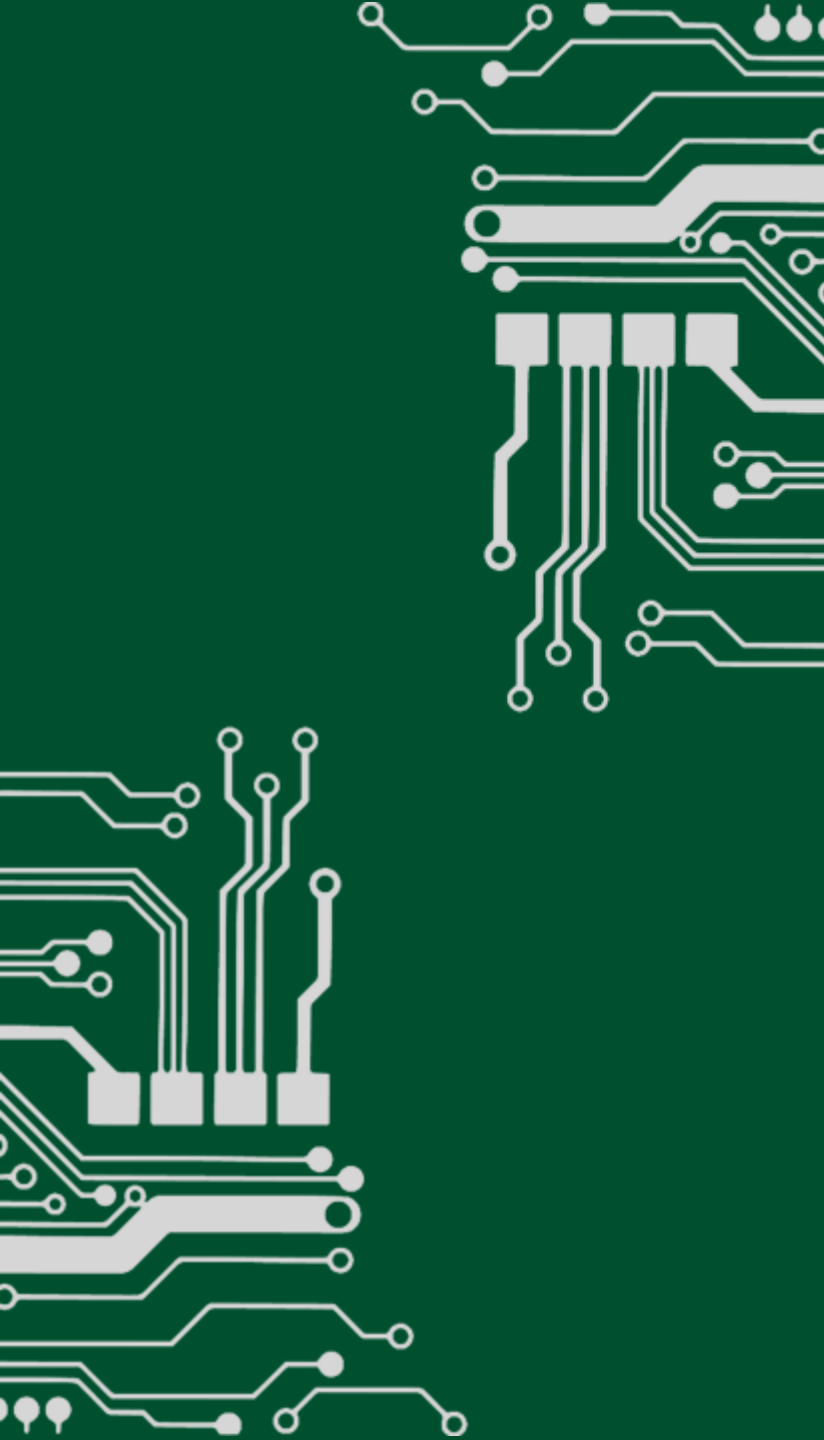
Research Computing Team Leader

IT Services



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Aire HPC Cluster



Overview of Aire



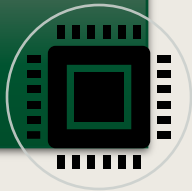
CPU systems	• 9,072 Cores • 52 Standard Nodes • 2 x 2.3TB High-Memory Nodes
GPU systems	• 84 x NVIDIA L40S GPUs • 28 Systems
Storage	• 138TB Lustre NVMe (Flash) • 3.6PB Lustre Scratch • 106TB Home Storage
Network	• 100 Gb/s OmniPath • 25GbE Ethernet

ARC3 -> AIRE	
CPU Cores	1.5x
Memory Bandwidth	6x
Max High Memory	3x
GPU Flops	28x
Scratch Storage	4.5x
NVMe Storage	New 138TB
Low Latency Network	1.8x

Nodes Technical Details

- Dell R6625 servers
- AMD Dual 84 core 2.2GHz (9634 Genoa-X)
- 8736 cores total
- 768GB DDR5-4800 Memory
- Dual 480GB M2 drives

CPU nodes
(52)



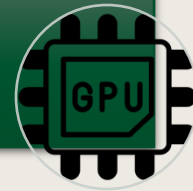
- AMD Dual 84 core 2.2GHz (9634 Genoa-X)
- 336 cores total
- 2.3TB DDR5-4800 Memory
- Dual 480GB M2 drives

High-Memory
nodes (2)



- Dell R7615 servers
- AMD 24 core 2.9GHz (9254 Genoa-X)
- 3 x NVIDIA L40S 48GB GPUs (PCIe)
- 84 GPU cards total
- 256GB DDR5-4800 Memory
- Dual 480GB M2 drives

GPU nodes
(28)



- AMD EPYC 9354P 32-Core Processor
- 192GB DDR5-4800 Memory
- 1 x NVIDIA A2 GPU

Login nodes
(4)



Storage & Quotas

Storage Type	Path, Symlinks	Associated Variable	Quota	Backup	Notes
Home Folder	/users/USERNAME ~	\$HOME	30GB capacity, 1-million files	Yes	Available all the time. Best for small, persistent files and user applications.
Scratch on Lustre (disk-based)	/mnt/scratch/USERNAME /scratch/USERNAME	\$SCRATCH	1TB capacity, 1.5-million files	No	High-performance storage ideal for large files or datasets that are actively being processed. Not backed up, so transfer important data elsewhere after processing.
Flash on Lustre (NVME-based)	/mnt/flash/tmp/job.JOB-ID /flash/tmp/job.JOB-ID	\$TMP_SHARED	1TB capacity, 1.5-million files	No	High-performance storage for temporary data during job execution. Available only via Slurm submission and purged when the job ends. Use for I/O-intensive tasks, e.g., tasks that involve frequent or large-volume read and write operations to and from storage devices like ML/AI. NVMe storage reduces latency and increases data transfer rates, minimizing bottlenecks, ensuring the compute nodes spend less time waiting for data.
Scratch on Computing Nodes	/tmp/job.JOB-ID	\$TMPDIR, \$TMP_LOCAL	None, subject to node storage availability	No	Node-local storage for job-specific temporary files. Access is fastest here but capacity is limited and data is lost once the job ends. Best for single-node jobs requiring quick read/write.

Modules (2nd April)

Compilers

- cuda
- gcc
- intel oneapi suit
- java

Libraries

- fftw
- hdf5
- lapack
- netcdf (c, fortran)
- openblas
- openmpi
- pytorch

Tools

- apptainer
- pixi
- spack
- texlive

Interpreters

- julia
- miniforge
- python
- R

Applications

- ambertools
- ansys
- castep
- crest
- gromacs
- imb
- matlab
- openfoam
- paraview
- rclone
- relion
- stata
- vtk
- xtb

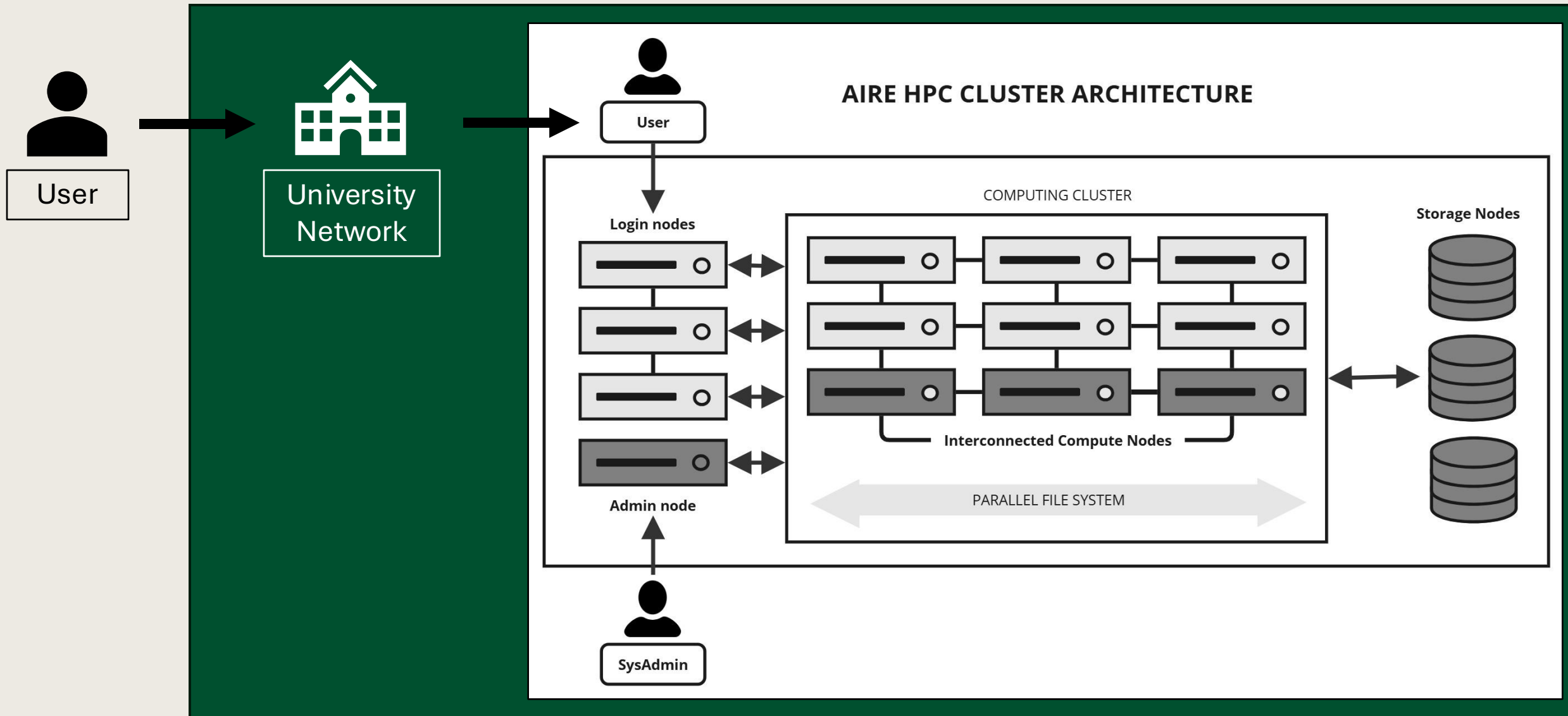


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HPC

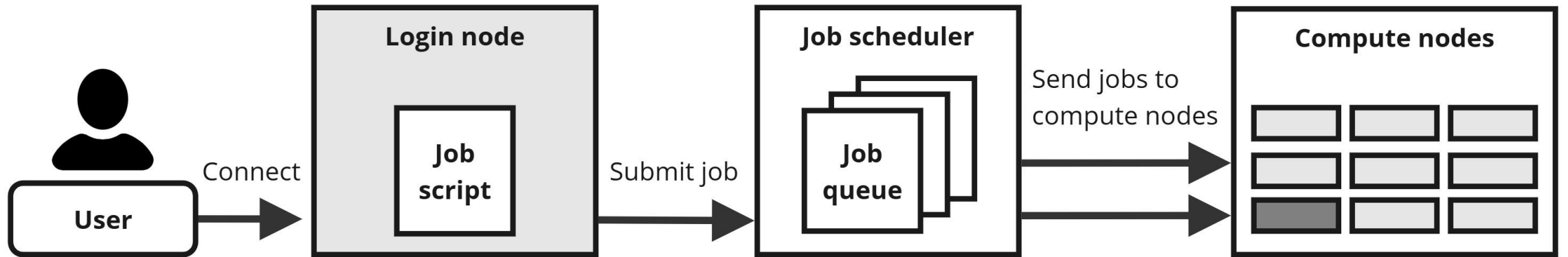
A (very brief) Overview

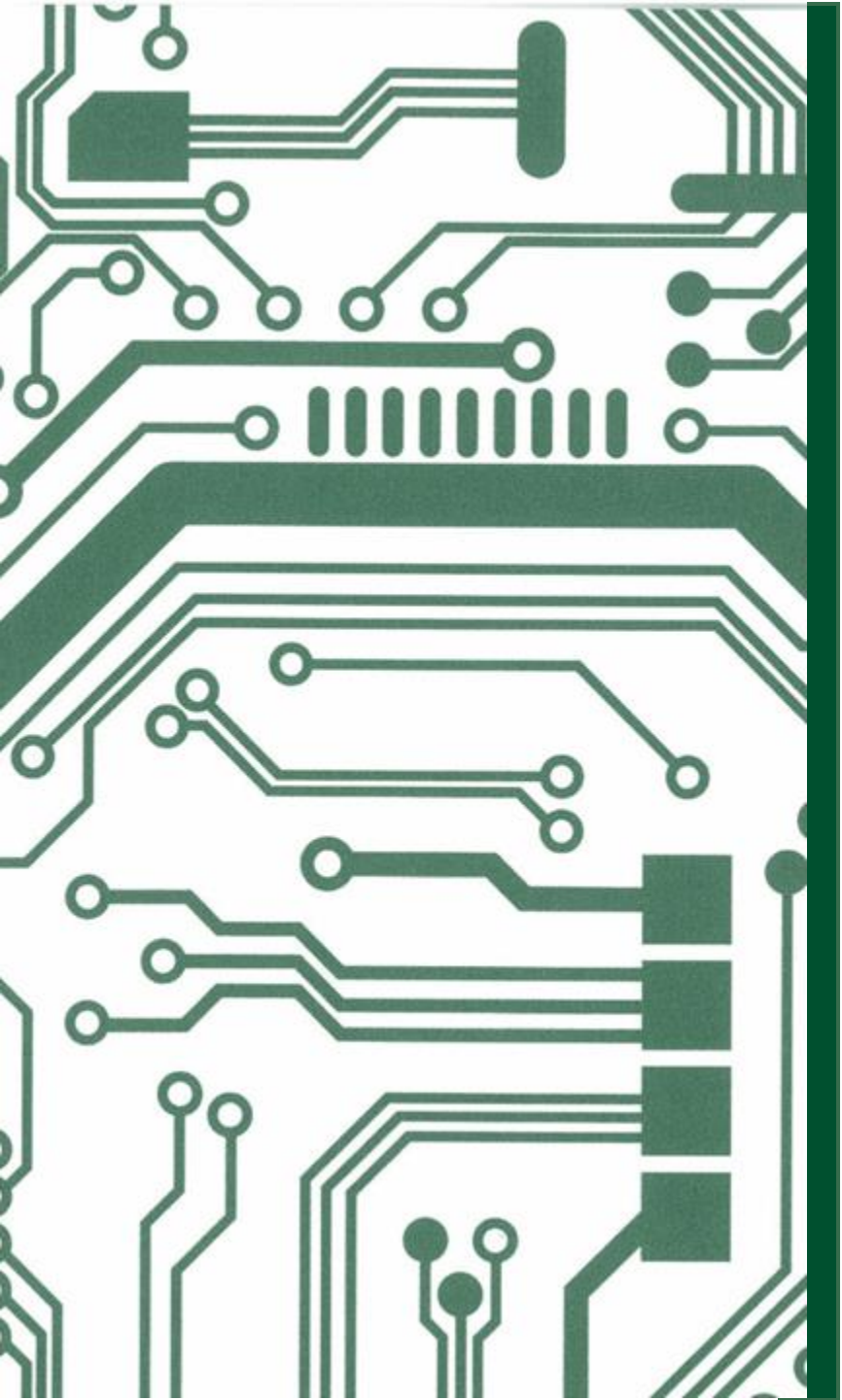
HPC architecture



Scheduler - SLURM

JOB SCHEDULING SYSTEM





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FAQ

Can I have an HPC account?

- All Postgraduate Researchers (academics, post-doctoral researchers and postgraduate students) are entitled to ARC accounts.
- Taught course students should speak to their supervisor before submitting an application request as they will be required to sponsor your application.
- You can request an account via the [IT service form](#).

Can I increase my quota?

- Yes, if there is a reasonable justification and the increase is temporary.
- Please be aware that we will exchange some messages to make sure that all options to improve the workflow and use of HPC have been taken before approving the request.
- You can submit your quota increase request via the [Research IT Query form](#)

The software I want is not available, can I request it to be installed?

- Yes, using the [HPC Application Install form](#).
- There are some requirements for new software to be installed centrally
 - Software version and maintenance
 - How many people will benefit from a central install
 - License
 - Team availability >> We currently have a large backlog and a long waiting time

What to do if my software was not installed centrally

- Users can install their own software into their home directory.
- There are a variety of approaches to this task. Installing from source code, using conda, spack, container.
- Training material for HPC installs: <https://arctraining.github.io/hpc2-software/>
- Support for user installs is limited.

Useful links

High Performance Computing (HPC) Account Request Form

- https://it.leeds.ac.uk/it?id=sc_cat_item&sys_id=4c002dd70f235f00a82247ece1050ebc

Aire Documentation

- <https://arcdocs.leeds.ac.uk/aire>

Research Computing Website

- <https://arc.leeds.ac.uk/>

HPC Application Install Form

- https://it.leeds.ac.uk/it?id=sc_cat_item&sys_id=48d5a6d70f275f00a82247ece1050ea0

Course Material: Installing and Managing Applications on the HPC

- <https://arctraining.github.io/hpc2-software/>

Available Courses

- <https://arc.leeds.ac.uk/courses/>

Research IT Query

- https://it.leeds.ac.uk/it?id=sc_cat_item&sys_id=7587b2530f675f00a82247ece1050eda

Questions?

Contact us at:

<https://arc.leeds.ac.uk/> & rcteam@leeds.ac.uk

